

A nontrivial weakly amenable Beurling algebra on a noncommutative group

Full answer to the existence challenge in arXiv:1702.06605

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Status. Candidate full solution, likely valid, pending human review. The example has an unbounded polynomial weight and a genuinely noncommutative convolution algebra.

1 Source challenge

Shepelska and Zhang write in the introduction of [3], printed page 3 (PDF page 3), that weak amenability of noncommutative Beurling algebras was still unclear and that they had not seen a nontrivial weakly amenable example which was noncommutative. This restates the first open problem in Zhang’s earlier paper [4].

The relevant sentence is:

“So far we have not even seen a non-trivial example of weakly amenable Beurling algebra which is not commutative.”

We give such an example explicitly.

2 Definitions

For a discrete group G and a weight $\omega : G \rightarrow [1, \infty)$ satisfying $\omega(xy) \leq \omega(x)\omega(y)$, the Beurling algebra $\ell^1(G, \omega)$ is the convolution algebra with norm

$$\|f\|_\omega = \sum_{x \in G} |f(x)|\omega(x).$$

A Banach algebra A is *weakly amenable* if every bounded derivation $A \rightarrow A^*$ is inner. It is *amenable* if every bounded derivation from A into the dual of every Banach A -bimodule is inner.

We write $A \hat{\otimes} B$ for the projective tensor product Banach algebra. The dual module conventions are

$$\langle c, a \cdot \Phi \rangle = \langle ca, \Phi \rangle, \quad \langle c, \Phi \cdot a \rangle = \langle ac, \Phi \rangle.$$

3 Proof intuition

The construction adds a finite noncommutative direction to a familiar commutative weakly amenable Beurling algebra. The main point is to prove that this does not create a new derivation.

For a derivation on $A\widehat{\otimes}B$, amenability of B first removes its restriction to the B -factor by subtracting one inner derivation. The remaining restriction to A takes values in the functionals which are central for B . Because A is commutative, that space is a symmetric A -module. A short duality argument shows that unital commutative weak amenability kills derivations into every such module. Hence the original derivation is inner.

Taking $A = \ell^1(\mathbb{Z}, (1 + |n|)^\alpha)$ and $B = \ell^1(S_3)$ completes the example: the first algebra is weakly amenable for $\alpha < 1/2$, while the second is amenable by its explicit finite-group diagonal.

N. Gronbaek showed in [10] that the weighted group algebra $L^1(G, \omega)$ is amenable if and only if G is an amenable group and ω is a diagonally bounded weight, i.e., the function $\omega(x)\omega(x^{-1})$ is bounded on G . The latter conditions actually imply that the weight ω is bounded up to a multiplicative factor. Hence, a nontrivial weighted group algebra is intrinsically not an amenable Banach algebra.

Weak amenability for commutative Banach algebras was introduced by Bade, Curtis, and Dales in [2]. Based on a characterization result of [2], Johnson later called a general Banach algebra A *weakly amenable* if every continuous derivation from A into A^* is inner. He showed in [15] that $L^1(G)$ is weakly amenable for all locally compact groups G .

Weak amenability of Beurling algebras has been studied by many authors. In [2] it was shown that $L^1(\mathbb{Z}, \omega_\alpha)$ for the additive group $\mathbb{Z}$ and the polynomial weight $\omega_\alpha(x) = (1 + |x|)^\alpha$ is weakly amenable if and only if $0 \leq \alpha < 1/2$. The same conclusion holds if $\mathbb{Z}$ is replaced with $\mathbb{R}$ ([5] [25] [30]). In [11] N. Gronbaek showed that the Beurling algebra of a commutative discrete group G is weakly amenable if and only if every non-trivial group homomorphism $\Phi: G \rightarrow \mathbb{C}$ satisfies

$$(1) \quad \sup_{g \in G} \frac{|\Phi(g)|}{\omega(g)\omega(g^{-1})} = \infty.$$

It turns out that this characterization is still valid for a general commutative locally compact group.

Theorem 1.1. [30, Theorem 3.1] *Let G be an Abelian locally compact group, and ω be a weight on G . The Beurling algebra $L^1(G, \omega)$ is weakly amenable if and only if (1) holds for every continuous non-zero group homomorphism $\Phi: G \rightarrow \mathbb{C}$.*

However, condition (1) is far from being sufficient for $L^1(G, \omega)$ to be weakly amenable if the group G is not commutative. A counterexample associated to discrete $SL_2(\mathbb{R})$ was obtained in [4]. In [26] the first author showed that with a non-trivial polynomial weight ω_α the algebra $\ell^1(\mathbb{F}_2, \omega_\alpha)$ is never weakly amenable. This contrasts with the results on commutative groups $\mathbb{Z}$ and $\mathbb{R}$ mentioned above. Similar investigations concerning the discrete $\mathbf{ax} + \mathbf{b}$ group were also conducted there. Overall, weak amenability of a non-commutative Beurling algebra is still very unclear. So far we have not even seen a non-trivial example of weakly amenable Beurling algebra which is not commutative. The related problem of weak amenability of the center algebra of a Beurling algebra has been studied in [1] [27] [30].

In this paper, in Section 2 we first characterize continuous derivations from $L^1(G, \omega)$ into its dual in terms of certain functions from $L^\infty(G \times G, \frac{1}{\omega \times \omega})$. We then show that the necessity part of Theorem 1.1 remains true if G is an IN group, improving a result of [30]. We further establish a criterion that rules out weak amenability of a Beurling algebra. As an application, we show that the weighted

Figure 1: The source context and existence challenge, rendered from `source_paper.pdf`, PDF page 3.

4 The tensor theorem

We first record the elementary module lemma needed in the proof.

Lemma 1. *Let A be a unital commutative weakly amenable Banach algebra. If X is a unital symmetric Banach A -bimodule, then every bounded derivation $\delta : A \rightarrow X$ is zero.*

Proof. Fix $\lambda \in X^*$. Define $T_\lambda : A \rightarrow A^*$ by

$$\langle T_\lambda(a), c \rangle = \langle \lambda, c \cdot \delta(a) \rangle \quad (a, c \in A). \quad (1)$$

Since X is symmetric, for $a, b, c \in A$ we have

$$\begin{aligned} \langle T_\lambda(ab), c \rangle &= \langle \lambda, c \cdot (a \cdot \delta(b) + \delta(a) \cdot b) \rangle \\ &= \langle \lambda, (ca) \cdot \delta(b) + (cb) \cdot \delta(a) \rangle \\ &= \langle a \cdot T_\lambda(b) + T_\lambda(a) \cdot b, c \rangle. \end{aligned}$$

Thus T_λ is a bounded derivation $A \rightarrow A^*$. Weak amenability makes it inner; since A is commutative, every inner derivation into A^* is zero. Putting $c = 1_A$ in (1) gives $\langle \lambda, \delta(a) \rangle = 0$. This holds for every $\lambda \in X^*$, so $\delta(a) = 0$ for every a . $\square$

Theorem 1. *Let A be a unital commutative weakly amenable Banach algebra and let B be a unital amenable Banach algebra. Then $A \widehat{\otimes} B$ is weakly amenable.*

Proof. Put $C = A \widehat{\otimes} B$ and let $D : C \rightarrow C^*$ be a bounded derivation. Regard A and B as the commuting closed subalgebras $A \otimes 1_B$ and $1_A \otimes B$ of C .

The restriction

$$b \mapsto D(1_A \otimes b)$$

is a derivation from B into the dual B -bimodule C^* . Amenability of B supplies $\Phi \in C^*$ such that this restriction agrees with the inner derivation $c \mapsto c \cdot \Phi - \Phi \cdot c$. Subtract that inner derivation from D . It is enough to prove that the resulting derivation D_0 is zero, and by construction

$$D_0(1_A \otimes b) = 0 \quad (b \in B). \quad (2)$$

Let

$$X = \{\Psi \in C^* : (1_A \otimes b) \cdot \Psi = \Psi \cdot (1_A \otimes b) \text{ for every } b \in B\}.$$

This is a closed unital A -submodule of C^* . It is symmetric as an A -module because $A \otimes 1_B$ is central in C : commutativity of A gives

$$(a \otimes 1_B)c = c(a \otimes 1_B) \quad (a \in A, c \in C).$$

Define $\delta : A \rightarrow C^*$ by $\delta(a) = D_0(a \otimes 1_B)$. The two factors commute, so applying D_0 to $(a \otimes 1_B)(1_A \otimes b) = (1_A \otimes b)(a \otimes 1_B)$ and using (2) yields

$$\delta(a) \cdot (1_A \otimes b) = (1_A \otimes b) \cdot \delta(a).$$

Hence δ is a bounded derivation $A \rightarrow X$. Lemma 1 gives $\delta = 0$.

Finally, on every elementary tensor,

$$D_0(a \otimes b) = D_0((a \otimes 1_B)(1_A \otimes b)) = 0.$$

Elementary tensors are dense in C , so $D_0 = 0$. Therefore D is inner, as required. $\square$

5 Full explicit answer

Corollary 1. *For every $0 < \alpha < 1/2$, let*

$$G = \mathbb{Z} \times S_3, \quad \Omega(n, \sigma) = (1 + |n|)^\alpha.$$

Then G is noncommutative, Ω is an unbounded symmetric weight, and the Beurling algebra $\ell^1(G, \Omega)$ is weakly amenable and noncommutative. In particular it is a nontrivial weakly amenable Beurling algebra of the type requested in [3, 4].

Proof. The inequality

$$1 + |m + n| \leq (1 + |m|)(1 + |n|)$$

shows that Ω is a weight. It is symmetric and unbounded, so it is not equivalent to the constant weight. The group G is noncommutative because S_3 is noncommutative.

Set

$$A = \ell^1(\mathbb{Z}, \omega_\alpha), \quad \omega_\alpha(n) = (1 + |n|)^\alpha, \quad B = \ell^1(S_3).$$

Bade–Curtis–Dales proved that A is weakly amenable exactly when $0 \leq \alpha < 1/2$ [1]; this is also recalled explicitly in the source paper [3, p. 3]. The finite group algebra B is amenable. Indeed,

$$M = \frac{1}{|S_3|} \sum_{\sigma \in S_3} \delta_\sigma \otimes \delta_{\sigma^{-1}} \in B \widehat{\otimes} B$$

is a diagonal: $b \cdot M = M \cdot b$ for $b \in B$ and the multiplication map sends M to δ_e . Thus B is even contractible.

The canonical convolution map gives the isometric Banach-algebra identification

$$A \widehat{\otimes} B \cong \ell^1(\mathbb{Z} \times S_3, \Omega).$$

Theorem 1 now gives weak amenability. Finally, convolution is noncommutative because the point masses supported on two noncommuting elements of S_3 do not commute. $\square$

The example is also a standard polynomial-weight example rather than merely an arbitrary inflation: for any finite generating set of $\mathbb{Z} \times S_3$, its word length differs from $|n|$ by a uniformly bounded amount in the finite S_3 direction. Thus Ω is equivalent to the corresponding usual polynomial word-length weight.

6 Verification and scope

The proof has no computational or unproved dependency. Its main checkable points are:

1. amenability of B is used with the dual B -module C^* , exactly as allowed by the definition;
2. after the B -restriction is killed, commutation of the two tensor factors forces the A -restriction into the B -central submodule;
3. centrality of $A \otimes 1_B$ makes that module symmetric;
4. Lemma 1 is proved directly from the defining $A \rightarrow A^*$ form of weak amenability;
5. Ω is unbounded and the algebra is genuinely noncommutative.

The result fully answers the source’s existence challenge. It does not settle the source’s broader tensor-product question asking whether the projective tensor product of *arbitrary* weakly amenable Beurling algebras must be weakly amenable. Theorem 1 gives a rigorous positive class: one factor may be any unital amenable algebra, provided the other is unital, commutative, and weakly amenable.

7 Bounded novelty check

On 9 August 2026 we searched the run’s registry, solution, attempt, and proof-gap indexes; the local arXiv source corpus; exact phrases from the source challenge; and web searches for noncommutative weakly amenable Beurling algebras and weak amenability of projective tensor products. No independent verified source for Corollary 1 was found.

A later arXiv preprint of Mehdipour and Rejali [2], Corollary 4.3, states a much broader pullback conclusion which would imply noncommutative examples. Its proof depends on Proposition 4.1(i). There, from injectivity of a pullback map $\mathfrak{S} : Q(G_2) \rightarrow Q(G_1)$ and the inclusion $\mathfrak{S}(I(G_2, \omega)) \subseteq I(G_1, \check{\omega})$, the argument asserts that an arbitrary non-inner quasi-additive function upstairs has the form $\mathfrak{S}(p_2)$. This is a surjectivity assertion not supplied by the preceding hypotheses or proof. Consequently that preprint does not, without a repair, provide a verified duplicate. The construction here uses a different, complete amenable-factor argument. Novelty confidence is therefore moderate and remains subject to expert review.

8 Human-review recommendation

Review as a candidate full affirmative answer to the existence challenge. The main verifier focus should be Theorem 1, especially the passage to the B -central symmetric module and the slicing argument in Lemma 1. A secondary literature review should check whether this amenable-factor tensor lemma, or this exact $\mathbb{Z} \times S_3$ construction, has appeared elsewhere under a different formulation.

References

- [1] W. G. Bade, P. C. Curtis Jr., and H. G. Dales, *Amenability and weak amenability for Beurling and Lipschitz algebras*, Proc. London Math. Soc. (3) **55** (1987), 359–377.
- [2] M. J. Mehdipour and A. Rejali, *Weak amenability of weighted group algebras*, arXiv:2209.08346v1 (2022).
- [3] V. Shepelska and Y. Zhang, *Non-weakly amenable Beurling algebras*, arXiv:1702.06605 (2017); Indiana Univ. Math. J. **67** (2018), 119–150.
- [4] Y. Zhang, *Weak amenability of commutative Beurling algebras*, Proc. Amer. Math. Soc. **142** (2014), 1649–1661; arXiv:1207.5033.